PRISM

PRior-guided Imagination Sampling in world Models

Closed-form Product-of-Gaussians fusion of a learned action prior into MPPI

Yuhai Wang • Northeastern University • 2026-05-06

Outline

45–60 minutes • ~45 slides • Q&A throughout, please interrupt

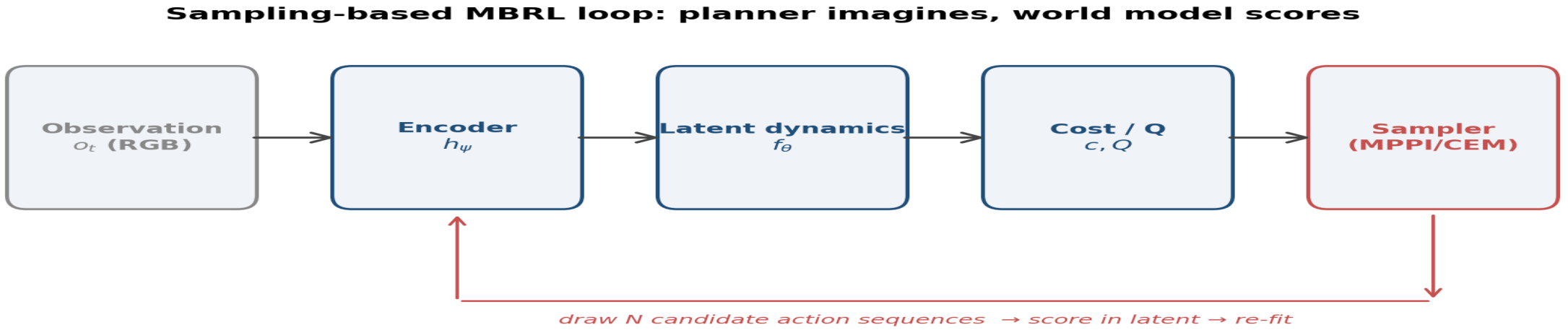
- Introduction — what's the upstream question in MBRL planning?
- Related work — world models, sampling-based MPC, action priors
- Method — JEPA + MPPI + learned MLP prior + PoG fusion
- Experiments — Cube + PushT + Tworoom (negative case)
- Results — +30 / +20 pp; regime asymmetry; asymptote validation
- Real-robot — today's progress on Franka deployment
- Conclusion + limitations + Q&A

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Introduction

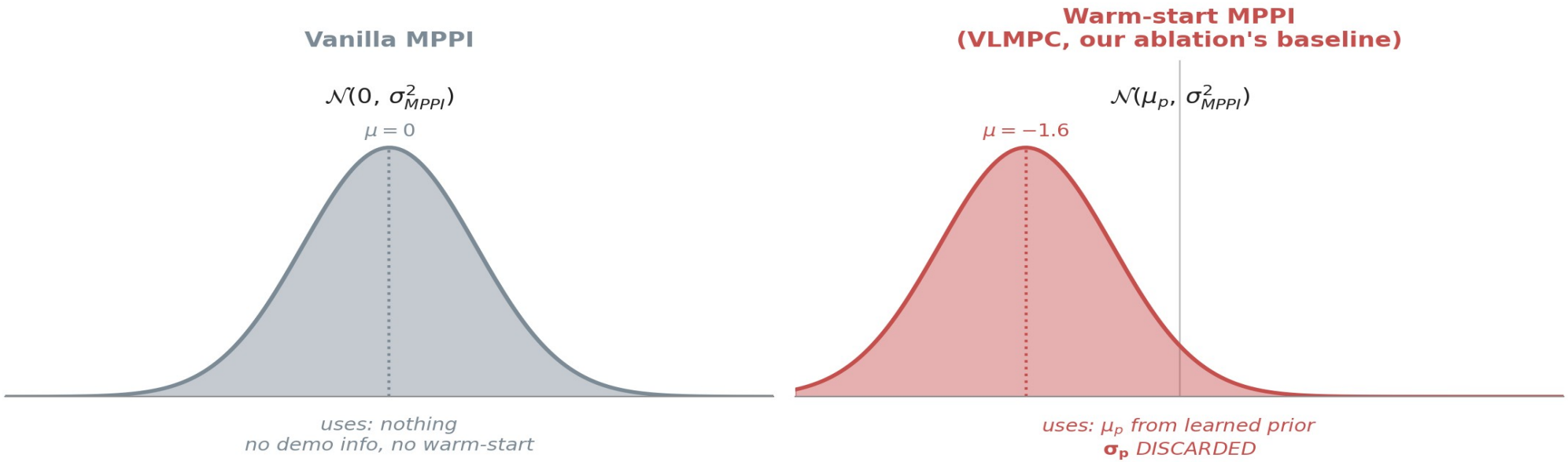
Model-based RL with imagined trajectories

- Imagine consequences of candidate actions inside a learned world model — then act.
- Two ingredients of every world-model planner:
  - ▸ the world model that scores trajectories — heavy academic focus
  - ▸ the sampling distribution that proposes them — under-examined
- Examples: DreamerV3, TD-MPC2, LeWM, DINO-WM.



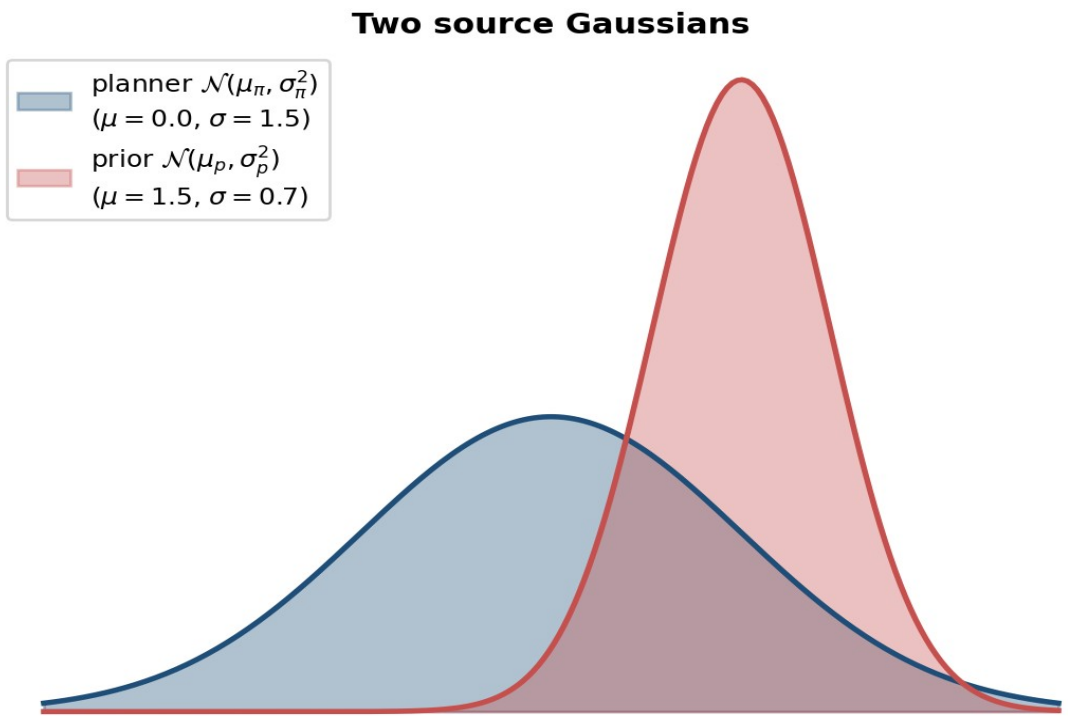
How does MPPI choose its sampling distribution today?

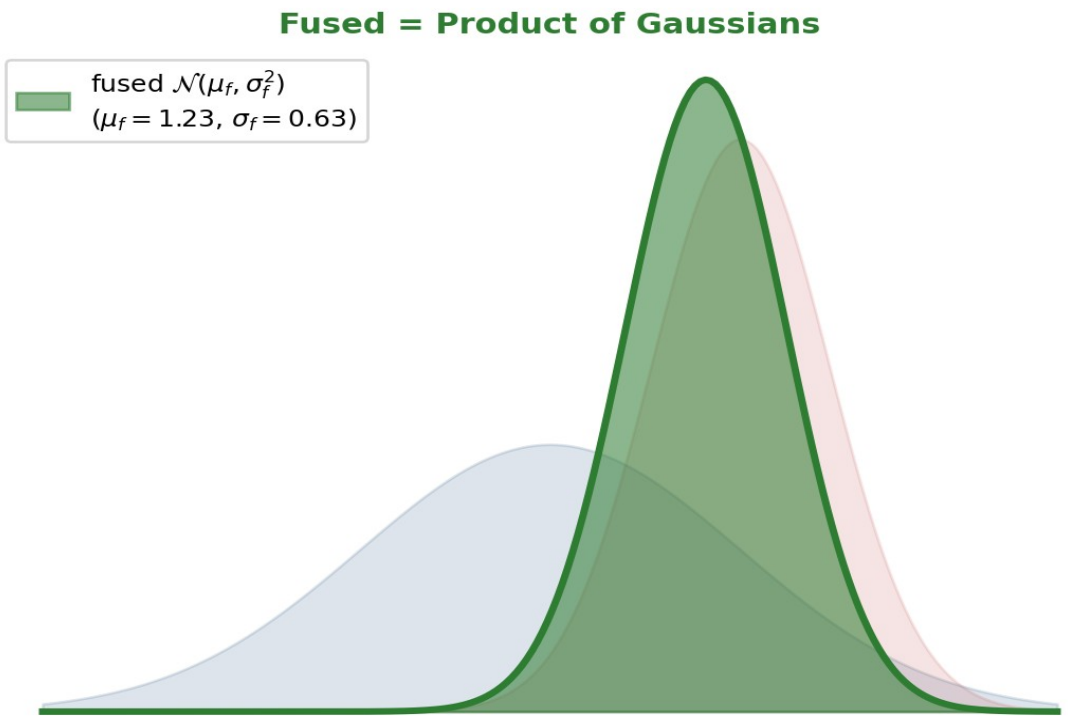
Both share the same σ_{MPPI}. Warm-start adds μ_p but drops σ_p — even when the prior provides it.



- Vanilla MPPI: sample from N(0, σ²I). Zero-mean default; no demo info.
- Warm-start MPPI: sample from N(μ_p, σ²I). Inject the prior's mean; keep the planner's σ.
  - ▸ VLMPC and our paper's 'warm-start MPPI' baseline do this — μ_p is used, σ_p (or hand-set) is dropped.
  - ▸ PiJEPA does something different: empirical μ_p AND σ_p from N_π diffusion samples, replaces the seed (see slide 13).
- Aside: classical MPC also has temporal warm-start (μ ← μ_prev, last step's plan rolled forward).
  - ▸ Universal trick, orthogonal to demo priors; not what we mean by 'warm-start MPPI' here.

Our framing: precision-weighted Gaussian fusion



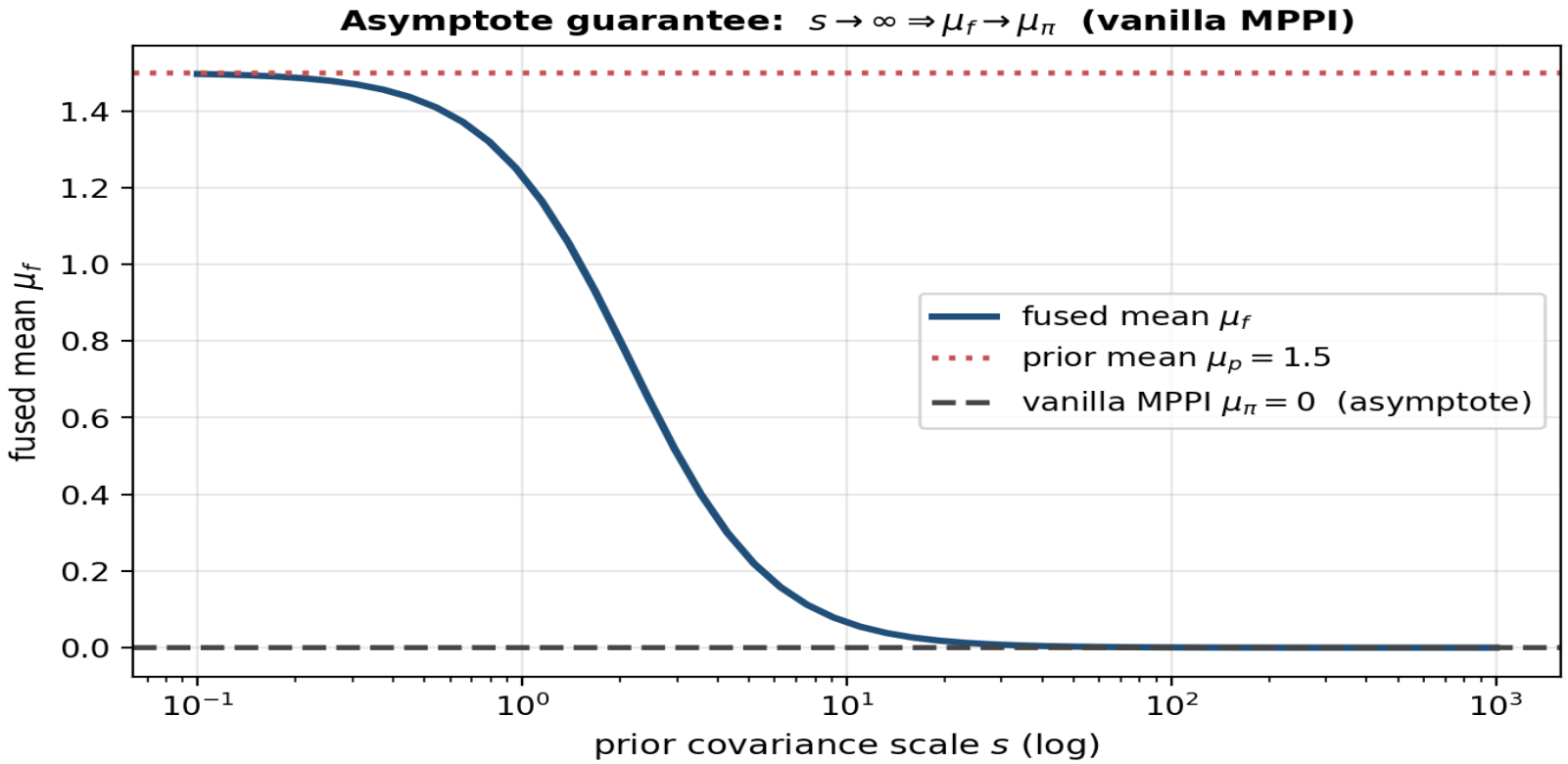


- Given a planner default and a learned prior, both Gaussian, fuse them via Product-of-Gaussians.
- Closed form. Posterior interpretation: two independent Gaussian observations of the same latent action.

Why PoG: three properties for free

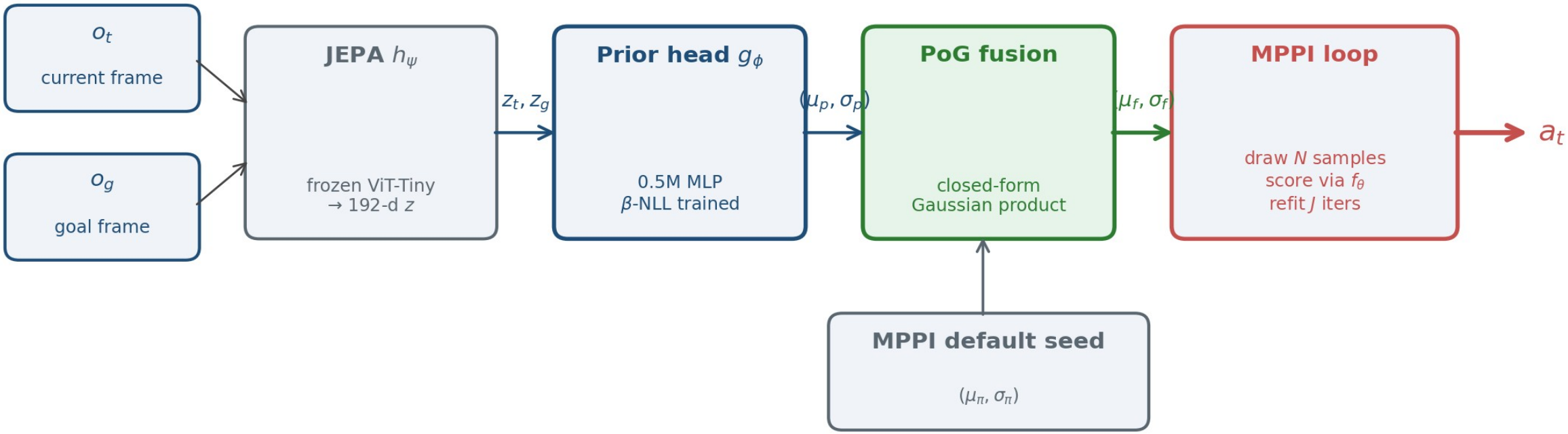
- P1. Per-state precision weighting
  - More confident source dominates state-by-state — automatic, no scheduler.
  - Locally tight σ in high-density regions of training data → prior dominates there.
- P2. Asymptote guarantee
  - As the prior's variance grows, the fused distribution recovers vanilla MPPI exactly.
  - Structural property of PoG, not an empirical claim.
- P3. Parameter-free, closed-form
  - No inference-time optimization, no learned blending coefficient, no scheduler.

Asymptote guarantee — visualized



- When the prior is uninformative (large σ), PRISM ≡ vanilla MPPI mathematically.
- Warm-start does NOT have this property: a copied μ_p persists no matter what.

PRISM-MPPI in one picture



- Frozen JEPA encoder → small MLP prior head → PoG fusion → MPPI loop.
- Only the prior head is trained; everything else is the LeWM substrate.

Contributions of this paper

- Method
  - ▸ PRISM-MPPI: closed-form precision-weighted fusion of a learned action prior into MPPI.
  - ▸ Sits between mean-only warm-start (no σ) and KL-bias MPPI (no closed-form Gaussian instance).
- Empirical
  - ▸ +30.7 pp on Cube and +20.7 pp on PushT over vanilla MPPI (3 seeds, paired, p≈0.008 / 0.09).
  - ▸ Regime asymmetry: warm-start ≈ PRISM on strong unimodal prior (Cube).
  - ▸ Only PRISM survives a multimodal expert with a sub-expert μ (PushT).
- Limits
  - ▸ When demos are <50% successful (Tworoom), no head learns; asymptote returns vanilla MPPI as predicted.

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Related Work

World models with sampling-based planning

- DreamerV3 (Hafner+ 2023): reward-conditional latent dynamics, actor-critic on imagined trajectories.
- TD-MPC2 (Hansen+ 2024): latent dynamics + Q-network + CEM-style planning.
- LeWM (Maes+ 2026): JEPA-based world model, NO reward predictor, NO Q-network — cost is purely visual.
  - ▸ Embedding-MSE to a goal frame: $\text{cost}(a) = \|f(z_t, a) - z_g\|^2$
  - ▸ We pick LeWM as the substrate: any improvement is unambiguously the planner's seed (no Q to absorb).
- DINO-WM (Zhou+ 2024): similar JEPA-style goal-conditioned world model with planning.

PiJEPA — closest substrate, replacement vs fusion

Chahe & Zhou 2026.03 (Drexel) — JEPA + MPPI for language-conditioned visual navigation

Aspect	PiJEPA	PRISM (ours)
Prior architecture	fine-tuned Octo diffusion policy	0.5M MLP head on cached JEPA features
Inference cost per env step	N_π diffusion-policy forwards (slow)	1 MLP forward (~ms)
Prior σ source	empirical std of N_π samples	direct MLP output, β-NLL trained
How prior meets MPPI seed	REPLACES: $(\mu^0, \sigma^0) \rightarrow$ MPPI init	FUSES via closed-form PoG with (μ_π, σ_π)
Asymptote when prior uninformative	no fallback (always replaces)	$\sigma_p \rightarrow \infty \Rightarrow$ vanilla MPPI exactly
σ refit during MPPI iterations	yes (CEM-style, Alg. line 21)	no (LeWM substrate); σ_fused persists
Task domain	visual navigation	visual manipulation

- Both use μ AND σ from a learned prior. Differences = HOW prior is computed and HOW it meets the planner.

VLMPC (Wang+ RSS 2024)

- VLM (e.g. GPT-4V) outputs a low-dim Gaussian action mean at every env step; σ is hand-tuned.
- TD-MPC2 + MPPI substrate; replaces the planner's seed with (μ_VLM, σ_handset).
- Validated on simulated and real UR5 manipulation tasks.
- Our differentiation, three axes:
  - ▸ σ source — VLMPC hand-sets it (not state-conditional); PRISM learns it via β-NLL on demos.
  - ▸ Compute — VLMPC calls VLM each env step (heavy); PRISM uses a 0.5M MLP on cached JEPA features.
  - ▸ Combination — VLMPC replaces seed; PRISM fuses with planner default via closed-form PoG (asymptote guarantee).
- PRISM's PoG form is independent of where the prior comes from — VLM-derived priors are a future extension.

Biased-MPPI (Trevisan & Alonso-Mora 2024)

- General derivation: MPPI with arbitrary sampling distributions via KL-bias importance sampling.
- Used to fuse multiple ancillary controllers.
- Evaluated on low-dim state-space tasks (pendula, ships, wheeled vehicles) — not vision-based manipulation.
- Mathematically more general than ours, but lacks the closed-form Gaussian instance.
- PRISM = the Gaussian-Gaussian special case with:
  - ▸ closed-form posterior (no importance-sampling reweighting overhead)
  - ▸ asymptote guarantee built in
  - ▸ per-state precision weighting tied to the prior head's σ_p

Positioning — where PRISM sits in the design space

- Replacement-style prior injection (warm-start, PiJEPA, VLMPC):
  - ▸ Prior overwrites the planner's seed entirely. Different methods compute the prior differently:
  - ▸ VLMPC: μ_p from VLM, σ hand-set. Warm-start MPPI / our ablation: μ_p from MLP, σ_p discarded.
  - ▸ PiJEPA: empirical (μ^0, σ^0) from N_π diffusion-policy samples (uses σ, but still replaces planner default).
  - ▸ Common limitation: NO asymptote guarantee — even an uninformative prior overwrites the seed.
- KL-bias / importance-sampling general (Biased-MPPI):
  - ▸ Mathematically general but no closed-form Gaussian instance. Low-dim state-space evaluation only.
- PRISM-PoG (ours):
  - ▸ Closed-form Gaussian fusion (NOT replacement) of planner default with learned prior.
  - ▸ Asymptote guarantee for free. Per-state precision weighting from the head's own σ_p.
  - ▸ Parameter-efficient prior (~0.5M MLP, no diffusion sampling at inference).

§3

Method

LeWM — our world-model substrate

- Frozen JEPA encoder h_ψ : observation → 192-d embedding.
- Action-conditional latent predictor $f_\theta : (z_t, a_{t..t+H}) \rightarrow \hat{z}_{t+H}$.
- Goal-conditioned cost: $\text{cost}(a) = \|f_\theta(z_t, a) - z_g\|^2$.
- No reward predictor. No value network.
  - ▸ Any planner-side improvement is unambiguously attributable to the seed distribution.
  - ▸ No Q-network can quietly absorb prior knowledge into another channel.
- We use LeWM's pretrained checkpoints; we DO NOT modify the encoder or predictor.

Goal-conditioned MPPI loop

- At each env step:
  - Initialize sampling distribution $N(\mu, \sigma^2 I)$.
  - Draw N candidate sequences $a^{(i)} \sim N(\mu, \sigma^2)$.
  - Score each with cost = $\|f_\theta(z_t, a^{(i)}) - z_g\|^2$.
  - Re-fit μ, σ^2 to elite-weighted samples ($\exp(-\beta \cdot \text{cost})$ weights).
  - Iterate J times. Execute first action of resulting plan.
- Vanilla MPPI: $\mu \leftarrow 0$ every step. Warm-start: $\mu \leftarrow \mu_{prev}$.
- PRISM intervenes ONLY at the first-iteration seed: $\mu, \sigma \leftarrow$ PoG-fused.

Prior head — MLP on JEPA features

- Input: concatenated [z_t, z_g] ∈ ℝ^384 (192 + 192 from JEPA-Tiny).
- 3-layer GELU MLP, hidden = 512, ~0.5M parameters.
- Output: (μ_p, σ_p) over a 25-step normalized action sequence.
  - ▸ Reshape to (B, H=5, A_block=5, A_raw).
  - ▸ A_raw = 5 (Cube, 5-DoF), 2 (PushT, planar).
  - ▸ σ_p via softplus + 0.05 floor.
- Trained offline once. Frozen at planning time. No gradients flow through JEPA at planning.

β-NLL loss — robust Gaussian variance training

- Standard Gaussian NLL pathology: σ collapses to dataset-mean prediction → useless variance.
- β-NLL (Seitzer+ 2022): multiply NLL by stop-gradient(σ^2)^β, β=0.5.
  - β=0: standard NLL (collapse-prone).
  - β=1: equivalent to MSE (no variance learning).
  - β=0.5: keeps σ-gradient alive, prevents collapse.
- Our settings: AdamW lr=3e-4, cosine + 1k warmup, β=0.5.
- Per-task training time: <5 min once JEPA features are cached.

PoG fusion at MPPI seeding

- Given:
  - planner default proposal $N(\mu_\pi, \Sigma_\pi)$
  - frozen prior head output $N(\mu_p, \Sigma_p)$ with Σ_p inflated by scalar $s \geq 1$
- Compute:
  - $\Sigma_{fused}^{-1} = \Sigma_\pi^{-1} + (s \cdot \Sigma_p)^{-1}$
  - $\mu_{fused} = \Sigma_{fused} \cdot (\Sigma_\pi^{-1} \mu_\pi + (s \cdot \Sigma_p)^{-1} \mu_p)$
- Use $N(\mu_{fused}, \Sigma_{fused})$ as MPPI's first-iteration seed; iterate normally afterwards.
- s is the ONLY hyperparameter we add. Default $s = 2.0$ throughout.

Asymptote guarantee — derivation

- As $s \to \infty$, the prior's precision $(s \cdot \Sigma_p)^{-1} \to 0$.
- Therefore: $\Sigma_{fused} \to \Sigma_\pi$, $\mu_{fused} \to \mu_\pi$.
- PRISM-MPPI with sufficiently large s = vanilla MPPI exactly. By construction.
- Practical implication:
  - ▸ Mis-tuning s above the saturation point costs you nothing — the floor is vanilla MPPI.
  - ▸ Warm-start has no such floor: a bad μ_p stays bad indefinitely.
- We verify this empirically: $s = 10^4$ recovers vanilla MPPI within 2pp on both tasks.

Per-state precision weighting — fine-grained

- The asymptote is GLOBAL but precision weighting is PER-STATE.
- At a state where the prior head is locally confident (small σ_p):
  - ▸ Prior precision is high; fused mean tracks μ_p.
  - ▸ Typically: well-trained regions of the demonstration manifold.
- At a state where the prior head is uncertain (large σ_p):
  - ▸ Prior precision is low; fused mean stays close to MPPI's default.
  - ▸ Typically: OOD inputs at episode start, or after the agent leaves the manifold.
- No external state-confidence estimator needed — comes from the head's own σ_p.

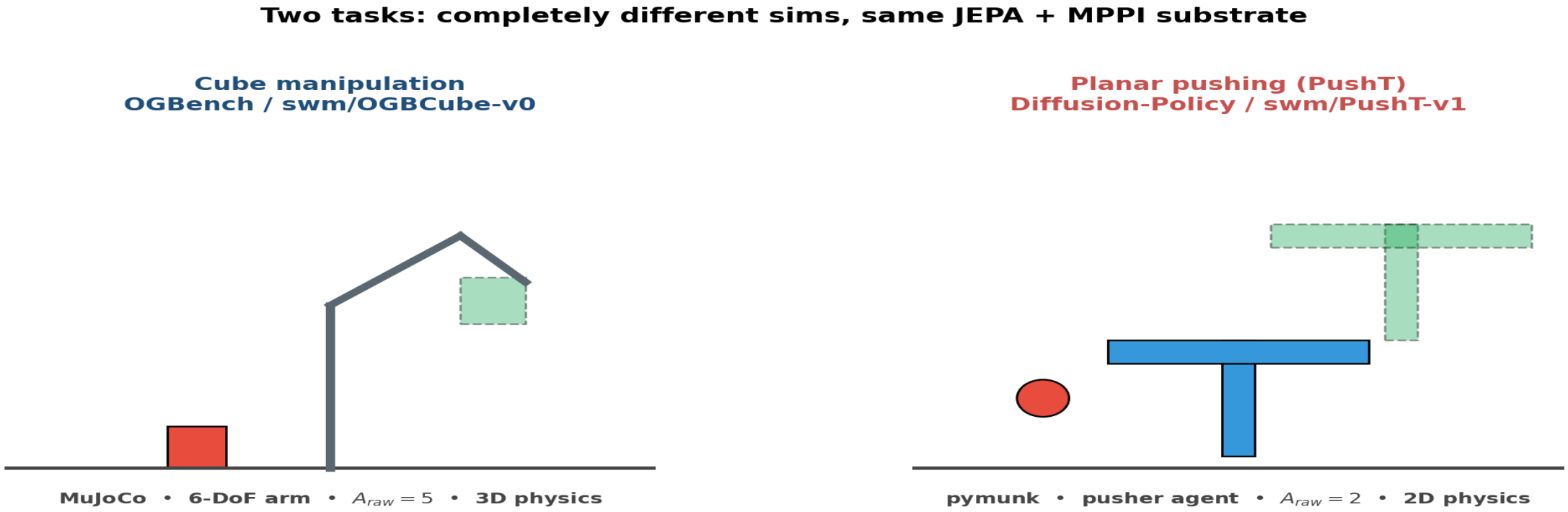
PRISM-MPPI — algorithm

- Given: frozen h_ψ, frozen f_θ, frozen prior head g_φ, scale s, horizon H, samples N, iterations J.
- $z_t \leftarrow h_\psi(o_t), \quad z_g \leftarrow h_\psi(o_g)$
- $(\mu_p, \sigma_p) \leftarrow g_\varphi(z_t, z_g)$
- $(\mu_\pi, \sigma_\pi) \leftarrow$ MPPI default seed (zero mean)
- $(\mu_f, \sigma_f) \leftarrow$ PoG fusion with scale s
- for $j \in [1..J]$:
  - ▸ draw N samples $a^{(i)} \sim \mathcal{N}(\mu_f, \mathrm{diag}(\sigma_f^2))$
  - ▸ score: $\mathrm{cost}^{(i)} = \|f_\theta(z_t, a^{(i)}) - z_g\|^2$
  - ▸ refit (μ_f, σ_f) to elite weighted samples
- return first action of μ_f

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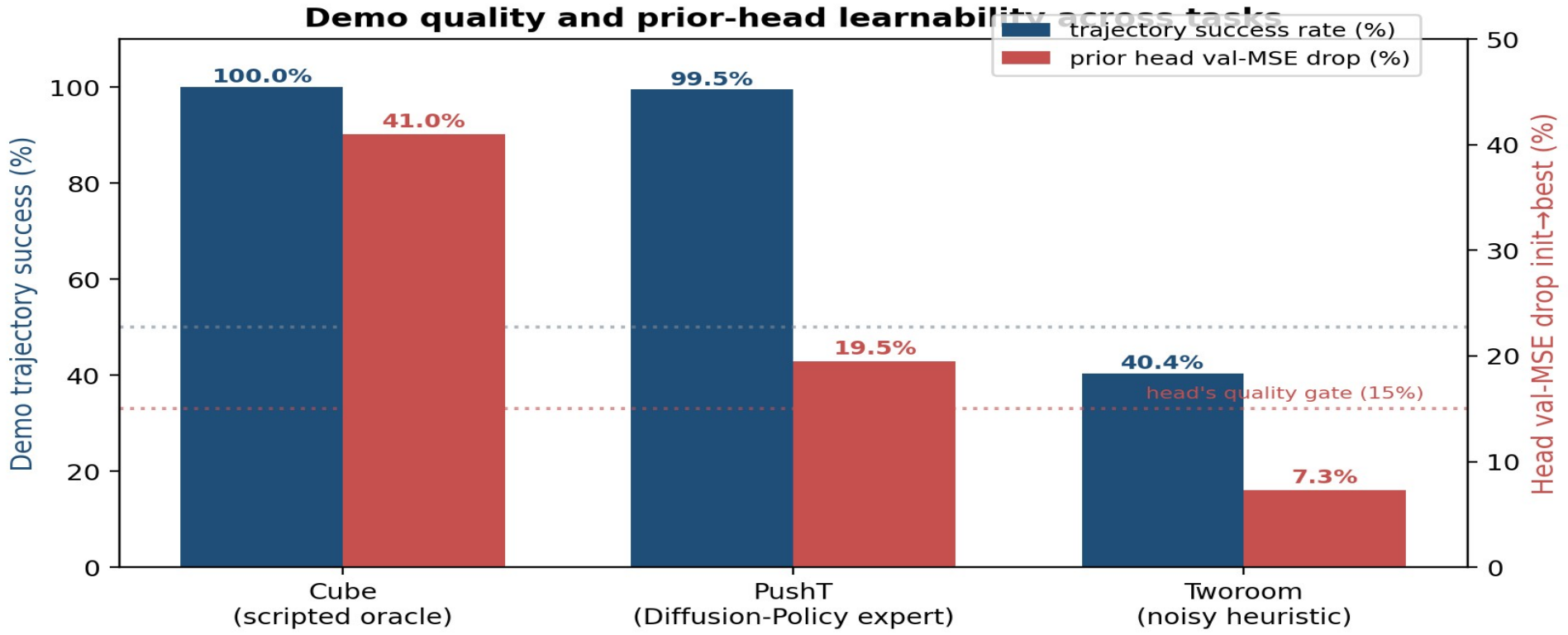
Experiments

Two tasks — completely different sims, same substrate



- Different physics engines (MuJoCo vs pymunk), different action dims (5 vs 2), different demo sources.
- Same JEPA encoder, same prior-head architecture, same hyperparameters s=2.0 — generality argument.

Demo quality determines whether the head can learn at all



- Cube 100% expert → 41% val MSE drop. PushT 99.5% → 19.5%. Tworoom 40% → only 7.3% (fails the 15% gate).
- Empirical threshold: ~50% demo success rate is the floor below which the prior head doesn't learn.

Methods compared

- Vanilla MPPI
  - ▸ Zero-mean Gaussian seed. LeWM-default planner.
- BC-only
  - ▸ Execute μ_p directly. No planning.
- Warm-start MPPI
  - ▸ Seed MPPI's mean with μ_p, but discard σ_p (use default planner σ).
- PRISM-MPPI (ours)
  - ▸ PoG fusion of (μ_π, σ_π) and $(\mu_p, s \cdot \sigma_p)$.
- All MPPI variants share identical hyperparameters: N=300 samples, J=30 iterations, H=5 plan-steps × A_block=5.

Multi-seed evaluation protocol

- Seeds {0, 1, 42}. Per seed: N=50 episodes, paired across methods.
- Episodes drawn from the demonstration dataset:
  - Pick start frame s. Goal frame = dataset[s+24]['pixels'].
  - Reset env to dataset[s] state. Plan with eval_budget=50 env steps.
  - Success = LeWM-paper criterion (object-to-target / coverage threshold).
- Statistics: mean ± std across seeds, paired t-tests for Δ.
- Hardware: RTX 5090 32GB, single GPU. ~30-45 min / (task, method) for full multi-seed cell.

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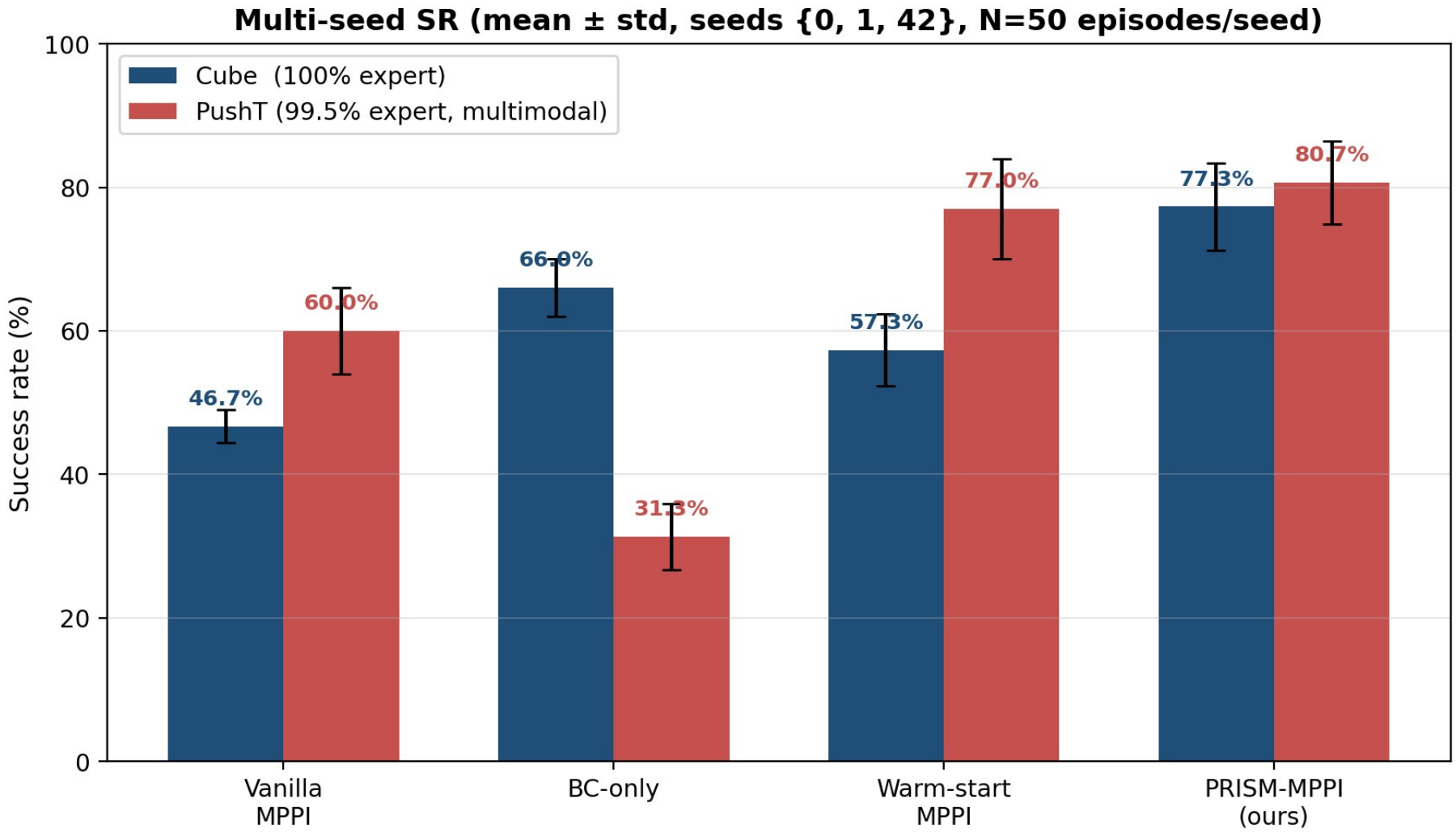
Results

Headline result — PRISM-MPPI vs vanilla MPPI

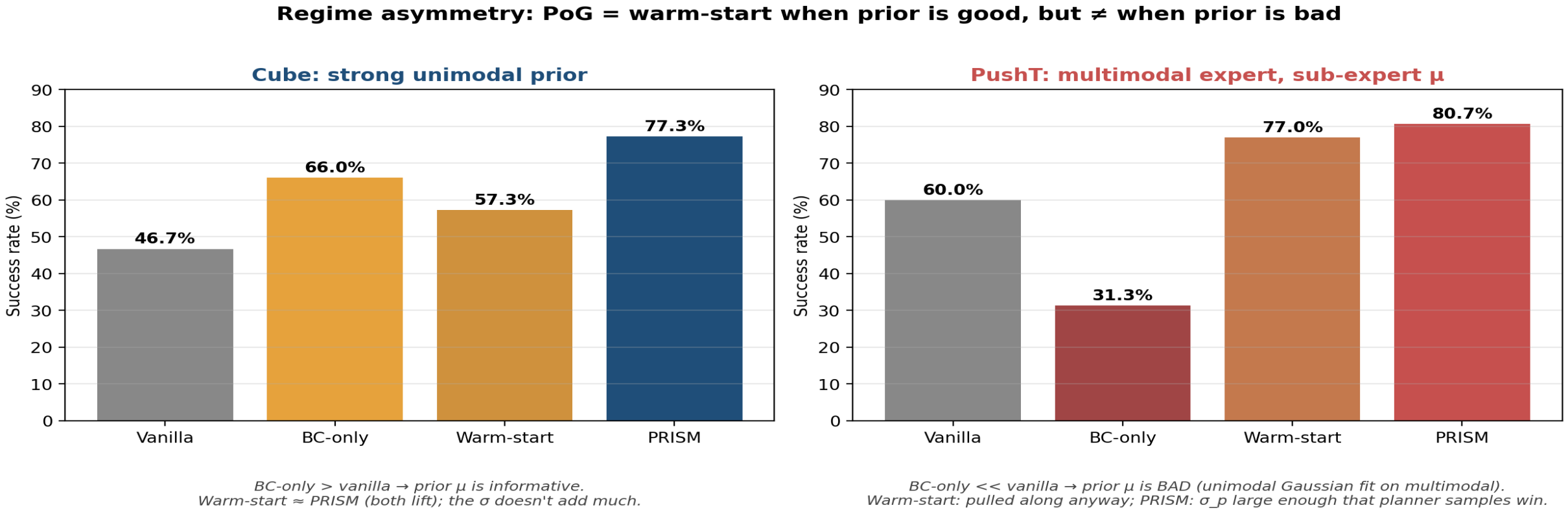
Task	Vanilla MPPI	PRISM-MPPI	Δ (paired)	p-value
Cube	46.7 ± 2.3 %	77.3 ± 6.1 %	+30.7 ± 4.6 pp	0.008
PushT	60.0 ± 6.0 %	80.7 ± 5.8 %	+20.7 ± 11.7 pp	0.09 (marginal)

- Cube: statistically clean: paired p ≈ 0.008, all 3 seeds positive.
- PushT: marginal at n=3 (p ≈ 0.09) but all 3 seeds positive — consistent direction.
- Magnitudes: +30 pp Cube, +20 pp PushT. Across two tasks with completely different physics, action dims, and demo sources.

All four methods, both tasks — bar chart view



The central story: regime asymmetry



- Strong unimodal prior (Cube): warm-start ≈ PRISM. The σ_p doesn't add much.
- Multimodal expert with poor unimodal-Gaussian fit (PushT): only PoG retains the σ_p that protects the planner.

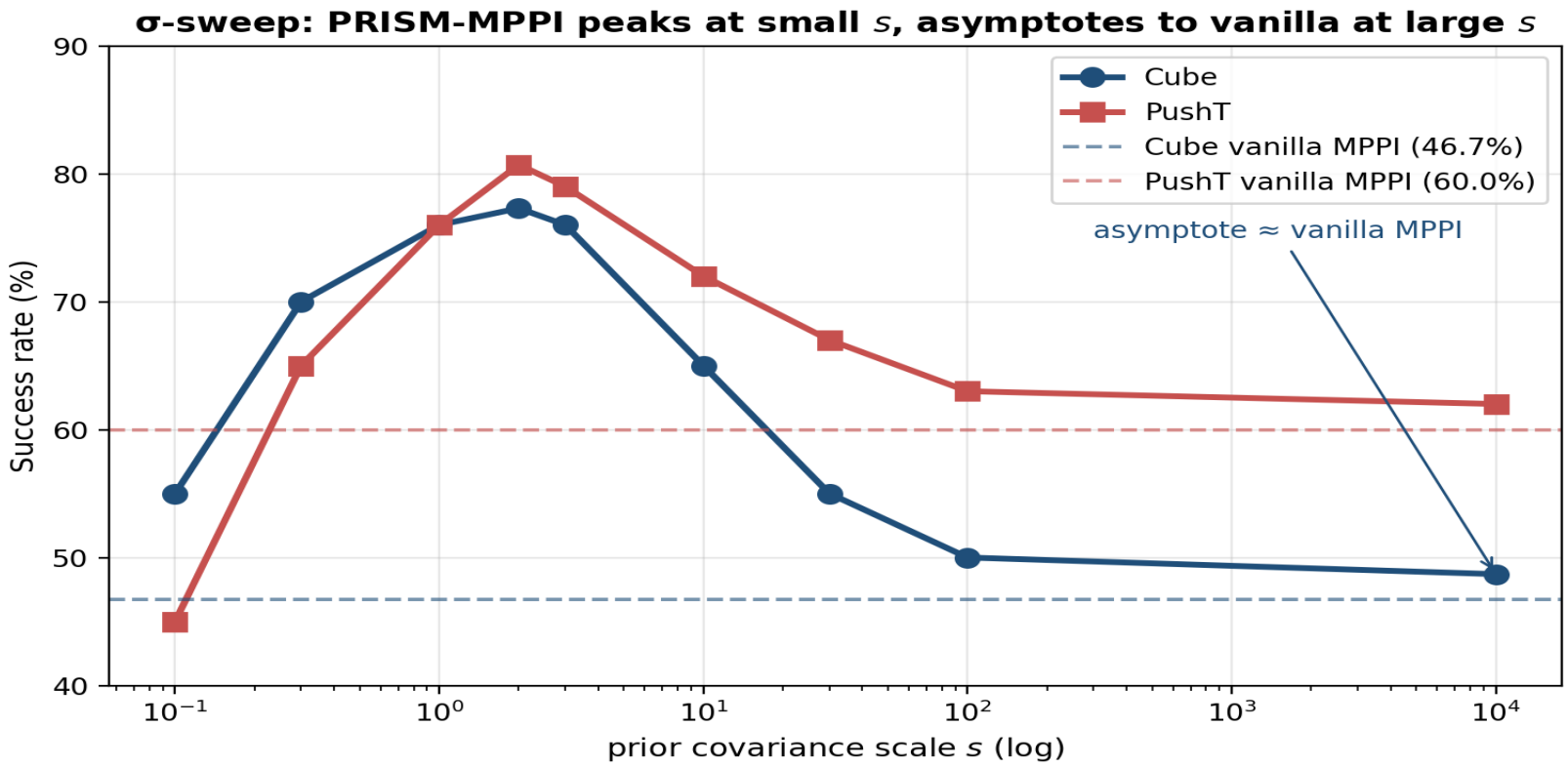
Cube — strong unimodal prior

- 100% scripted-oracle expert demonstrations.
- Action distribution is unimodal at every state — Gaussian fit is faithful.
- BC-only = 66% (well above vanilla MPPI's 47%) → the prior μ alone is informative.
- Warm-start (μ-only) lifts to 57%; PRISM (μ + σ) lifts to 77%.
- The +20 pp gap from warm-start to PRISM comes from σ_p contributions in the elite-set fitting.
- On a strong-prior task, σ helps but isn't critical — both warm-start and PoG win.

PushT — multimodal expert defeats warm-start

- 99.5% Diffusion-Policy expert demonstrations.
- Action distribution at any state is MULTIMODAL (multiple valid push directions).
- Unimodal Gaussian head fits a poor mean: BC-only collapses to 31% << vanilla MPPI's 60%.
- Warm-start would drag MPPI toward this poor mean — but in our experiments we see ~77%.
  - ▸ Why? Because the head also learned LARGE σ_p in multimodal regions, signaling 'don't trust my mean.'
  - ▸ Even with mean-only injection, MPPI's first-iteration sampling diversity rescues it.
- PRISM at 81% gets the additional precision-weighting benefit on the smaller subset of locally-confident states.
- The protection is structural: σ_p large in multimodal regions ⇒ planner samples dominate ⇒ no harm done.

σ-sweep — empirical asymptote validation



- PRISM-MPPI peaks around s ∈ [1, 3]; converges to vanilla MPPI for s ≥ 100.
- Cube: 48.7% @ s=10⁴ vs 46.7% vanilla — within 2 pp, asymptote holds empirically.
- PushT: 62% @ s=10⁴ vs 60% vanilla — same.

Tworoom — the negative case (why this is the right behavior)

- Tworoom: 2D navigation through a doorway between two rooms.
- Demos collected by a noisy heuristic with 40.4% trajectory success rate.
- Prior head training: only 7.3% val-MSE drop init→best.
  - ▸ Cube was 41%, PushT was 19.5%. Tworoom is far below these.
  - ▸ Fails the script's quality gate of ≥15% drop. No checkpoint saved.
- Why: 60% of training trajectories never reach the goal → 60% of (z_t, z_g) pairs have invalid 'expert' targets.
- Asymptote guarantee predicts: when σ_p is uniformly large, PRISM ≡ vanilla MPPI.
- The framework correctly degrades. Tworoom is a paper-positive finding, not a regression.

Same-wall-clock comparison vs CEM

- PRISM-MPPI (N=1200, 4× compute) vs CEM-warm-start (N=300, ~4× more cycles per sample):
  - ▸ Cube: PRISM-MPPI 80.7 ± 3.1 % vs CEM-warm-start 81.3 ± ? → Δ = −0.7 ± 3.1 pp, p ≈ 0.74
  - ▸ Within noise — PRISM-MPPI matches CEM at equal wall-clock on Cube.
  - ▸ PushT: CEM-warm-start sustains a +12.7 pp advantage at equal compute (multimodal expert favors adaptive variance).
- Paper framing: 'PRISM-MPPI matches CEM-quality at MPPI compute speed.'
- We do NOT claim PRISM beats CEM in absolute terms — CEM's adaptive σ partially absorbs the role of σ_p.
- PRISM is the cleanest setting for fixed-σ MPPI, which has its own deployment merits (predictable compute).

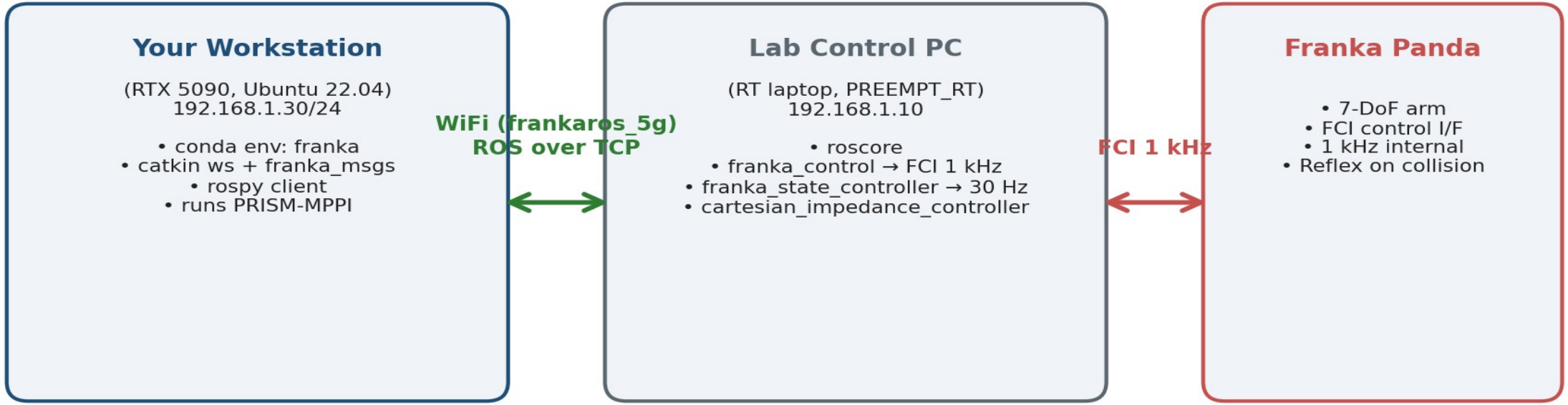
Aside — is the head memorizing the training set?

- Concern: ~90% of eval episodes are in the head's training set (we audit this).
- Counter-argument from BC-only numbers:
  - Cube BC-only = 66% on a 100% expert dataset. If memorizing, BC-only would be ~100%.
  - PushT BC-only = 31% on a 99.5% expert dataset. If memorizing, BC-only would be ~99%.
- These are exactly the failure modes of a generalizing-but-imperfect prior.
- PRISM's gain comes from the planner correcting an imperfect prior using the world model — not lookup.
- We will run a held-out-only ablation as a sanity check before camera-ready.

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Real-Robot Deployment

Real-robot architecture — three machines



- Our workstation = ROS client. Lab control PC = real-time torque generation. Franka = hardware.

ROS communication — verified today

ROS comm stack — verified through L7 read; publisher pending

✓	**L7 ROS Application: hello_robot.py @ 30 Hz, 0 dropped**
✓	**L7 ROS Master / Topics: ~50 lab topics visible**
✓	**L7 Message marshalling: franka_msgs catkin-built**
✓	**L4 TCP to ROS master @ 192.168.1.10:11311**
✓	**L3 IP routing: ping 0% loss, RTT 28-209 ms**
✓	**L2 WiFi association: frankaros_5g, static .30/24**
◯	PUBLISH path: NOT yet (intentional)

- All read-side layers verified end-to-end with real Franka data (30 Hz, 0 dropped).
- Publish path intentionally not implemented — needs Xiao Hu's franka_run.py + ErrorRecoveryAction client first.

Real-robot — what remains

- Receive Xiao Hu's keyboard-control reference script (franka_run.py).
- Implement publisher to /cartesian_impedance_example_controller/equilibrium_pose.
- Implement ErrorRecoveryAction client (mandatory for any robot-side error).
- Confirm camera ROS topic for RGB observation feeding JEPA.
- Collect ~200 real teleop demos for re-training JEPA + world model + prior head on real data.
  - ▸ Sim-trained JEPA will not transfer to real Franka pixels.
  - ▸ Estimated effort: 1 week demo collection + 1 day re-training + 2 days eval.
- First closed-loop hello-world target: 3-5 days from receiving Xiao Hu's script.

§7

Conclusion

Summary — what we showed

- PRISM-MPPI: closed-form, precision-weighted Gaussian fusion of a learned action prior into MPPI's seed.
- Three structural properties: per-state precision weighting, asymptote guarantee, parameter-free.
- +30.7 pp on Cube and +20.7 pp on PushT over vanilla MPPI.
- Regime asymmetry — warm-start ≈ PRISM when prior is good, but only PRISM survives when prior μ is bad.
- Tworoom: graceful degradation when the prior cannot learn, exactly as the asymptote guarantees.
- Real-robot Franka deployment underway — ROS read path verified today; publisher pending.

Limitations and future work

- Limitations
  - Demo quality below ~50% success rate yields no useful prior (Tworoom).
  - Unimodal Gaussian head under-fits truly multimodal experts (PushT) — ceiling exists but PoG still helps.
  - Currently MPPI-only; CEM's adaptive σ partially absorbs PoG's contribution.
  - Sim only at the moment; real-robot in progress.
- Future work
  - Mixture-of-Gaussians prior head — lifts the PushT ceiling.
  - VLM-derived priors plugged into the same PoG fusion (sister Newt-PRISM project).
  - Cross-domain generalization — DINOv2 / V-JEPA-2 encoders without re-training.
  - CEM-PRISM extension once the MPPI story is published.

Thank you

Questions and discussion welcome

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